



Sizhang Xu

UI Engineering · Interactive Systems

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EDUCATION

University of Washington

M.S. in Technology Innovation · Global Innovation Exchange (GIX) · In progress

Sep 2026 – Jun 2028 (expected)

Tsinghua University

Graduate studies · GIX dual-degree program

Lab affiliation: PI Lab, Department of Computer Science and Technology

Sep 2025 – Jun 2028 (expected)

Shanghai Jiao Tong University

B.Eng. in Industrial Design · Rank 1/30 · National Scholarship

Semester exchange: University College London · Sep 2024 – Jan 2025

Jun 2021 – Jun 2025

INTERNSHIP EXPERIENCE

ByteDance | Seed

Feb – Jun 2026

LLM Data Product Intern | **LangGraph** Evaluation Agent

- **DataSeek post-evaluation analysis:** Developed and iterated a **Python/LangGraph** prototype that turned model-evaluation results into capability-level failure diagnoses and training-data recommendations. Led module requirements and analysis design through iterative releases; provided templates for four core analysis operators for engineering integration with Dataverse and R&D Agent orchestration.
- **Analysis pipeline design:** Designed an agent pipeline: reference reasoning → comparative error diagnosis → capability aggregation → training-data mapping. Used **best-of-N** reference sampling and LLM judging to establish a reasoning baseline; compared model and reference reasoning to extract question-level failure causes and capability labels. Aggregated fine-grained findings into mid-level capabilities for pretraining, SFT, and RL data plans. Designed a **four-quadrant analysis framework** (reference answer present/absent × scoring rubric present/absent) for objective and subjective evaluations.
- **Prompt system design:** Led the design of **30+** structured prompt templates with scenario-action capability names and constraints against unsupported capability attribution. To retain long-tail failure causes and make results comparable across evaluation sets, designed two-stage aggregation: first bucket by 18 predefined capability labels, then adapt grouping within each bucket to the number of failure causes.
- **Product design and iteration:** Independently wrote PRDs and led review rounds. Led manual quality reviews of **700+** reports, identified nearly 100 **P0/P1/P2** issues, and established issue-to-requirement tracking and fix follow-up. Designed short-term frontend fallbacks and longer-term prompt changes for recurring report failures.

Sohu | Mint Lab

Oct 2025 – Jan 2026

AI Product Management Intern

- **Knowledge-graph content creation:** Built an AI-assisted authoring tool that turned fragmented ideas into editable, traceable topic graphs, outlines, and drafts. Designed the workflow around creators' information overload and difficulty organizing material.
- **Prompt architecture and optimization:** Led prompt design across association → structuring → linear overview → drafting. Used **Root/Hyponym** constraints to organize root topics and child concepts; combined chain-of-thought prompting with self-ranking and voting to address uncontrolled expansion, hallucinations, and inconsistent topic hierarchies.
- **Engineering implementation:** Independently developed Python end-to-end test scripts. Reworked the generation pipeline by splitting title and description generation into sequential model calls. Designed an Outline-to-Draft flow so creators could edit structure before expanding it into prose and trace generated content back to its outline.
- **Evaluation system:** Built a **six-dimension** quality rubric with stage-specific weights and a four-level evaluation set from recurring content topics. Compared model combinations under quality, latency, and cost constraints; used regression cases to measure prompt changes and check for regressions across iterations.

INTERNSHIP EXPERIENCE · CONTINUED

Tsinghua University | Future Laboratory

Jun – Sep 2024

Human-Computer Interaction Research Intern

- **VR interaction research:** Reviewed **30+** papers and reports on cutting, chopping, and tearing in VR. Used findings on weak haptic immersion to define test scenarios and build a wearable force-feedback prototype.
- **Force-feedback modeling:** Modeled the relationship between motor input angle and output resistance. Developed three **C#/Unity** task modules for cutting fruit and vegetables, chopping a wooden target, and tearing paper.
- **User testing and iteration:** Tested three interaction tasks with **eight participants**. Examined the timing of virtual contact and physical feedback using signal traces; investigated mismatched feedback and compensatory effort near the end of cutting motions to guide interaction revisions.

PUBLICATIONS

“In Sync with Me”: A Robotic Body-Doubling Probe for ADHD Programmers’ Deep Work

Chenyang Li, **Sizhang Xu**, Hancheng Ye, and Ziyu Huang
UbiComp Companion '26 (Poster), 2026 · Accepted / in press

[DOI](#)

SELECTED PROJECTS

Financial Fables | Interactive Learning Web App

2026

JavaScript · HTML/CSS | xuxu8421.github.io/financial-fables

- Built and deployed a **15-story** JavaScript learning app with quizzes, hash-based deep links, **History API** navigation, and local-storage progress. Implemented keyboard navigation, semantic quiz controls, explanatory feedback, and an **aria-live** region.
- Supported reduced-motion preferences and mobile AVIF images with JPEG fallbacks. The 15 mobile assets totaled **0.84 MB** versus **3.99 MB** for desktop WebP assets, a **79%** byte reduction across different formats and dimensions.

Real-Time Sleep Research Platform

2026

Python · **FastAPI** · Vue/JavaScript · **WebSocket** · BLE

- Built a **Python/FastAPI** and Vue research platform connecting BLE sensors, audio inference, and intervention control. Implemented REST controls and **WebSocket** state updates for live signals, disconnected devices, stale readings, and cooldown status.
- Separated audio capture, inference, and BLE I/O with a **ring buffer**, worker thread, and **asyncio** loop. Implemented the intervention **state machine**, **JSONL** event logs, and background waveform compression and **NPZ** writes.

CAMPUS LEADERSHIP

SJTU Student Union (35th term) — Presidium Member

Coordinated liaison and communication across 34 schools.

School of Design Student Union, SJTU (6th term) — Executive President

Nanyang News Agency, SJTU — Presidium Member & Deputy Secretary-General

Managed content for the university’s official WeChat account and led planning for 80+ campus-culture features. Team results included 170,000+ new followers and 1.75M+ article reads.

SKILLS

Programming & tools: Python, C#, Figma, Cursor (AI-assisted development)

Engineering: JavaScript, HTML/CSS, Vue, Unity

English tests: TOEFL 108; CET-6 641; CET-4 657